INCH POUND

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SUPERSEDING
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DETAIL SPECIFICATION

STEEL BAR (LOW EXPANSION, FREE Cutting, 36 PERCENT NICKEL)

Inactive for new design after 20 December 1995

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a low coefficient of expansion, free cutting, steel bar.

1.2 Classification. Bars should be of the following finishes as specified (see 6.2):

Hot finished - Scale not removed
Cleaned - Pickled or blast cleaned
Centerless ground - Round bars only

2. APPLICABLE DOCUMENTS

2. 1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A751 - Standard Test Methos and Practices for Chemical Analysis of Steel Products
ASTM-E8/A8M - Standard Test Methods for Tension Testing of Metallic Materials
ASTM-E381 - Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings
ASTM-E1282 - Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods, Ores, and Related Materials

(Copies of this document are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9510

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SAE INTERNATIONAL (SAE)

SAE-AMS2806 - Identification Bars, Wire, Mechanical Tubing, and Extrusions, Carbon and Alloy Steels and Corrosion and Heat-Resistant Steels and Alloys

(Copies of these documents are available from www.sae.org or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The steel shall be manufactured by the crucible or the electric-furnace process.

3.1.1 Process. Unless otherwise specified, the steel bars shall be hot rolled or forged.

3.2 Chemical composition. The chemical composition shall conform to the composition of Table I.

TABLE I. Chemical composition.

Carbon Percent (max.)	Manganese Percent (max.)	Silicon Percent (max.)	Nickel percent	Selenium percent	Iron percent
0.15	1.00	0.35	35.0 - 36.5	0.15 - 0.25	Remainder 1/

1/ Incidental elements must not be present in such proportions as will modify the low expansivity characteristics, reduce the strength or add to the difficulty of machining.

3.3 Mechanical properties. The mechanical properties of the steel shall conform to the requirements of Table II.

TABLE II. Mechanical properties.

Tensile strength (min.)	Yield Strength (0.2% offset min.)	Elongation in 2 inches (min.)	Reduction of area (min.)
75,000 psi	45,000 psi	30 percent	40 percent

3.4 Finish. Hot finished bars shall be furnished as forged or rolled. Bars specified in the cleaned finish shall be free from scale. Centerless ground bars shall be bright and free from pits, scratches, or other injurious defects (see 1.2).

3.5 Macrostructure. Visual examination of deep acid etched bars shall show no more than slight pitting and no evidence of change in structure from the surface to the center (see 4.5).

3.6 Tolerances. Variations from normal dimensions shall be within the permissible limits shown in Tables III to VI.

3.7 Length. The bars shall be commercially straight and free from twist and damaged ends (see 6.2, 6.2.1).

3.7.1 Exact lengths. Bars may be ordered to exact lengths or in lengths expressed as a multiple of a definite unit. A tolerance of plus 0.125 inch will be allowed on material so ordered.

3.7.2 Mill lengths. Unless otherwise specified, material shall be furnished in full lengths of 6 to 20 feet. Not more than 10 percent of lengths shorter than 10 feet will be accepted.

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TABLE III. Hot-finished bars: rounds and squares.

Diameter or distance across flats	Permissible variations		Out of round or square 1/
	Plus	Minus	
Inches	Inch	Inch	Inch
0.313 and less	0.005	0.005	0.008
Over 0.313 to 0.438 inclusive	.006	.006	.009
Over 0.438 to 0.625 inclusive	.007	.007	.010
Over 0.625 to 0.875 inclusive	.008	.008	.012
Over 0.875 to 1.000 inclusive	.009	.009	.013
Over 1.000 to 1.125 inclusive	.010	.010	.015
Over 1.125 to 1.250 inclusive	.011	.011	.016
Over 1.250 to 1.375 inclusive	.012	.012	.018
Over 1.375 to 1.500 inclusive	.014	.014	.021
Over 1.500 to 2.000 inclusive	.016	.016	.023
Over 2.000 to 2.500 inclusive	.031	0	.023
Over 2.500 to 3.500 inclusive	.047	0	.035
Over 3.500 to 4.500 inclusive	.063	0	.046
Over 4.500 to 5.500 inclusive	.078	0	.058
Over 5.500 to 6.500 inclusive	.125	0	.070
Over 6.500 to 8.000 inclusive	.156	0	.085

1/ Out of round is the difference between the maximum and minimum diameters of the bar, measured at the same cross section. Out of square is the difference in the two dimensions at the same cross section of a square bar, each dimension being the distance between opposite faces, the adjacent sides forming approximately 90 degree angles.

TABLE IV. Hot-finished bars: hexagons and octagons.

Distance across flats	Permissible variations		Maximum difference 3 measurements (for hexagons only)
	Plus	Minus	
Inches	Inch	Inch	Inch
0.500 and less	0.007	0.007	0.011
Over 0.500 to 1.000 inclusive	.010	.010	.015
Over 1.000 to 1.500 inclusive	.021	.021	.025
Over 1.500 to 2.000 inclusive	.031	.031	.031
Over 2.000 to 2.500 inclusive	.047	.047	.047
Over 2.500 to 3.500 inclusive	.063	.063	.063

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TABLE V. Hot-finished bars: flats.

Ordered width	Permissible variations				
	Width		Thickness		
	Plus	Minus	0.125 inch to 0.500 inch thick inclusive (±)	Over 0.500 inch to 1.000 inch thick inclusive (±)	Over 1.000 inch to 2.000 inches thick inclusive (±)
Inches	Inch	Inch	Inch	Inch	Inch
1.000 and less	0.016	0.016	0.008	0.010	
Over 1.000 to 2.000 inclusive	.031	.031	.012	.015	0.031
Over 2.000 to 4.000 inclusive	.063	.031	.015	.020	.031
Over 4.000 to 6.000 inclusive	.094	.063	.015	.020	.031
Over 6.000 to 8.000 inclusive	.125	.156	.016	.025	.031
Over 8.000 to 10.000 inclusive	.156	.188	.021	.031	.047

TABLE VI. Centerless-ground bars: rounds only.

Diameter	Permissible variations	
	Plus	Minus
Inches	Inch	Inch
Less than 1.000	0.002	0.002
1.000 to 1.500 exclusive	.0025	.0025
1.500 to 4.000 inclusive	.003	.003

3.8 Identification of product. Each piece shall be identified in accordance with SAE-AMS2806, and the markings shall include the number of the heat of metal, the manufacturer's identification, condition, and designation of this specification.

3.8.1 Bars of the same size and smaller than 0.500 inch diameter of 0.375 inch width of flat shall be properly marked or tagged at each end and bundled. An extra marking or tag shall be attached to each bundle.

3.9 Workmanship. The steel bars furnished under this specification shall be uniform in quality and condition, smooth, free from pipes, laps, cracks, twists, seams, flakes, heat checks, slag hard spots, porosity, slivers, scabs, rolled-in scale, fissures, gas cavities, sponginess, excessive segregation, or any other defects which due to their nature, degree, or extent will be detrimental to the performance of parts fabricated from the metal.

4. VERIFICATION

4.1 Conformance inspection. Inspection of the steel is classified as conformance inspection (see 4.2 through 4.5.3).

4.2 Examinations.

4.2.1 Sampling for examination of dimensions, finish, workmanship and identification marking. Units of product shall be randomly selected in accordance with Table VII to represent each lot of material of one heat, the same nominal dimensions, and offered for acceptance at one time.

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TABLE VII. Sampling for examination of product.

Lot size	Sample size	Acceptance no.
1 to 4	1	0
5 to 110	5	0
111 to 500	7	0
501 to 800	10	0
801 to 1200	15	0
Over 1200	25	0

4.3 Chemical analysis.

4.3.1 Size of lot. For the purpose of sampling for chemical analysis a lot shall consist of all material made from the same heat or melt of steel. If the heat or melt cannot be identified, a lot shall be limited to 500 pounds of bars of each size and configuration.

4.3.2 Sampling. Unless otherwise specified, the sample shall represent material from the topmost portion of an ingot. If unable to identify the melt, a sample shall be taken from each of two bars in the lot and analyzed separately. Drillings may be taken from broken test specimens. Drillings shall be fine, clean, and free from oil, dirt, grit, surface scale, or other foreign matter, and shall consist of not less than 2 ounces.

4.3.3 Test method. Chemical analysis shall be made in accordance with ASTM-E1282 or ASTM-A751 on samples selected as specified in 4.3.2 to determine compliance with 3.2. In the event of disputes, the analysis of ASTM-E1282 shall be the basis for acceptance or rejection.

4.4 Tensile test.

4.4.1 Size of lot. For the purpose of sampling for mechanical tests a lot shall consist of all bars of the same size produced from the same heat or melt of steel and offered for delivery at the same time. If the heat or melt cannot be identified, a lot shall be limited to 500 pounds of bars of each size and configuration.

4.4.2 Sampling. Not less than 2 bars nor more than 2 percent of the bars in each lot shall be selected for tension test specimens. From each bar selected a single tension test specimen shall be prepared. Extra length shall be provided on bars from which test specimens are to be taken. Material shall be provided for possible extra tests. The test specimens should be located in that part of the bar nearest the uppermost part of the ingot as cast, unless otherwise specified.

4.4.3 Test method. The tensile test shall be performed in accordance with ASTM-E8/E8M on samples selected as specified in 4.4.2 to determine compliance with 3.3. Wherever practicable, bars shall be tested in full size, the cross section of the specimen being the full cross section of the material. If full size specimens are not practicable, the Standard 0.500 inch (12.5 mm) Round Tension Test Specimen with 2 inch (50 mm) Gage Length or the Standard Sheet – Type, 0.500 inch Wide Rectangular Tension Test Specimen shall be machined and tested as indicated in ASTM-E8/E8M. The Standard Round Tension Test Specimen shall be machined from any point at mid-radius of the bar.

4.5 Macroetch test.

4.5.1 Size of lot. For the purpose of sampling for macroetch examination, a lot shall consist of all bars of the same size produced from the same ingot of steel. In case it is impracticable to identify the material by ingots, a lot shall be limited to 500 pounds of bars of each size and configuration.

4.5.2 Sampling. Unless otherwise specified, a specimen for the macrotech test shall be taken from the uppermost end of the top bar in each ingot as cast. If unable to identify the ingot, specimens shall be taken from each of two bars in the lot.

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4.5.3 Test method. The macroetch test shall be performed in accordance with ASTM-E381 on samples selected as specified in 4.5.2. Specimens for the macroetch test shall consist of pieces approximately 0.500 inch thick cut from the end of the bar. The faces representing the cross-section of the bar shall be finished smooth and flat by a fine machine cut or by grinding. The etching solution shall be a 1 to1 solution of concentrated hydrochloric acid and water. The specimens shall be immersed in the solution at 160 ±3° F for 15 to 20 minutes. After etching, the specimens shall be washed in running water or steam and scrubbed to remove any cloaking deposits. The specimens shall then be examined for compliance with 3.5.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from then managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of machined parts in which it is necessary to have little change in size with temperature change. The low expansivity characteristics of this material are applicable over a limited range of temperatures. The most useful range is up to the inflection temperature which is at approximately 390° F. The mean coefficient of expansion between 32° F and the inflection temperature is approximately 1.4 x 10^{-6} per ° F. If a lower coefficient of expansion is necessary in the above temperature range, bars should be ordered cold drawn, annealed and quenched, or in special cases quenched and stabilized.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.
b. Finish (see 1.2 and 3.4).
c. Diameters of bars required.
d. Length of bars and whether "EXACT" or mill lengths are desired (see 3.7).
e. Packaging requirements (see 5.1).

6.2.1 Length. When bars are ordered in multiples of definite length, allowance should be made for possible waste in cutting.

6.2.2 Finish. In determining the surface finish to be specified, consideration should be given to whether contemplated operations will remove the surface finish, whether pickling by the using agency is contemplated, or if, for other reasons, pickling is deemed unnecessary and whether the bars are to be machined on "automatic" machines.

6.3 Subject term (key word) listing.

Bars
Free Machining Steels
Iron Alloys
Nickel Steels
Thermal Expansion

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Preparing activity:
DLA - CQ

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Review activities:
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